

Question Bank

Course Title: Microbiology and Infection control

Course Code: (MED12114)

Prepared by Dr/ Rania Hamed

Spring semester-2025/2026

Introduction to Microbiology and Infection Control

1. Microbiology is the study of:

- a. Large organisms
- b. Microscopic organisms
- c. Plants only
- d. Animals only

2. Bacteria are classified as:

- a. Multicellular eukaryotes
- b. Unicellular prokaryotes
- c. Acellular organisms
- d. Fungi

3. Viruses are considered:

- a. Living cells
- b. Prokaryotic organisms
- c. Non-living infectious agents
- d. Eukaryotic cells

4. Prokaryotic cells are characterized by:

- a. Presence of nucleus
- b. Absence of nucleus
- c. Presence of mitochondria
- d. Presence of Golgi apparatus

5. Cocci bacteria are:

- a. Rod-shaped
- b. Spiral-shaped
- c. Spherical
- d. Filamentous

6. Viruses are unable to reproduce independently because they:

- a. Lack DNA
- b. Lack metabolic machinery
- c. Are large in size
- d. Have a cell wall

7. The main difference between prokaryotic and eukaryotic cells is:

- a. Size
- b. Shape
- c. Presence of nucleus
- d. Color

8. Normal flora mainly functions to:

- a. Cause infection
- b. Protect against pathogens
- c. Destroy tissues
- d. Inhibit immunity

9. Fungi are classified as:

- a. Prokaryotic
- b. Acellular
- c. Eukaryotic
- d. Viral

10. Infection is defined as:

- a. Presence of microorganisms
- b. Multiplication of microorganisms causing disease
- c. Presence of bacteria only
- d. Presence of viruses only

11. Infection occurring after 48 hours of hospital admission is called:

- a. Community infection
- b. Nosocomial infection
- c. Chronic infection
- d. Latent infection

12. Failure of hand hygiene leads to:

- a. Increased immunity
- b. Infection transmission
- c. No effect
- d. Rapid recovery

13. Microorganisms that lack a true nucleus are:

- a. Fungi
- b. Protozoa
- c. Bacteria
- d. Viruses

14. Viruses enter host cells mainly to:

- a. Obtain energy
- b. Reproduce
- c. Move
- d. Grow

15. Personal protective equipment (PPE) is used to:

- a. Kill bacteria
- b. Prevent infection transmission
- c. Increase immunity
- d. Promote microbial growth

16. Bacteria differ from viruses in that bacteria:

- a. Are smaller
- b. Reproduce independently
- c. Lack DNA
- d. Have no structure

17. Normal flora and pathogens differ in that:

- a. Both are harmful
- b. Both are beneficial
- c. Normal flora are beneficial, pathogens are harmful
- d. There is no difference

18. Which of the following is absent in prokaryotic cells?

- a. Cytoplasm
- b. Ribosomes
- c. Nucleus
- d. Cell membrane

19. The most effective method for preventing hospital infections is:

- a. Antibiotics
- b. Hand hygiene
- c. Surgery
- d. Isolation

20. Infection control aims to:

- a. Increase infection
- b. Prevent the spread of infection
- c. Destroy tissues
- d. Reduce immunity

21. The branch of microbiology that studies fungi is:

- a. Bacteriology
- b. Virology
- c. Mycology
- d. Parasitology

22. Protozoa are classified as:

- a. Prokaryotes
- b. Acellular organisms
- c. Eukaryotic unicellular organisms
- d. Viruses

23. Bacterial DNA is found in:

- a. Nucleus
- b. Nucleoid
- c. Mitochondria
- d. Golgi apparatus

24. Viruses are classified as:

- a. Cellular organisms
- b. Prokaryotic cells
- c. Acellular particles
- d. Eukaryotic cells

25. Rod-shaped bacteria are called:

- a. Cocci
- b. Bacilli
- c. Spirals
- d. Filaments

26. Bacteria are prokaryotic because they:

- a. Have a nucleus
- b. Lack a nucleus
- c. Have mitochondria
- d. Have membrane-bound organelles

27. Ribosomes in bacteria are responsible for:

- a. Movement
- b. Protein synthesis
- c. DNA storage
- d. Protection

28. Eukaryotic cells are characterized by:

- a. Absence of nucleus
- b. Presence of nucleus
- c. Lack of organelles
- d. Simple structure

29. Normal flora may become pathogenic when they:

- a. Increase in number
- b. Change their location
- c. Improve immunity
- d. Remain constant

30. Viruses require host cells because they:

- a. Need oxygen
- b. Need nutrients
- c. Cannot reproduce independently
- d. Are large

31. Flagella in bacteria are used for:

- a. Nutrition
- b. Movement
- c. Protection
- d. Reproduction

32. Misuse of antibiotics leads to:

- a. Faster recovery
- b. Antibiotic resistance
- c. No effect
- d. Increased immunity

33. Contact with contaminated surfaces without precautions leads to:

- a. Sterilization
- b. Infection transmission
- c. Immunity
- d. Disinfection

34. Loss of normal flora may lead to:

- a. Immunity
- b. Opportunistic infection
- c. Healing
- d. Protection

35. Viral attachment to host cells occurs by:

- a. Ribosomes
- b. Spikes (peplomers)
- c. Cytoplasm
- d. Flagella

36. The distinguishing feature of eukaryotic cells is:

- a. Cytoplasm
- b. Ribosomes
- c. Nucleus
- d. Cell membrane

37. Bacteria differ from fungi in that bacteria are:

- a. Eukaryotic
- b. Acellular
- c. Prokaryotic
- d. Multicellular

38. Fungi are more complex than bacteria because they:

- a. Are smaller
- b. Lack DNA
- c. Have a nucleus and organelles
- d. Cannot reproduce

39. Effective infection control measures include:

- a. Antibiotic overuse
- b. Hand hygiene and PPE
- c. Ignoring infection
- d. Reducing staff

40. Normal flora protects the body by:

- a. Causing infection
- b. Competing with pathogens
- c. Destroying tissues
- d. Reducing immunity

Key Answers

1-b 2-b 3-c 4-b 5-c 6-b 7-c 8-b 9-c 10-b
11-b 12-b 13-c 14-b 15-b 16-b 17-c 18-c 19-b 20-b
21-c 22-c 23-b 24-c 25-b 26-b 27-b 28-b 29-b 30-c
31-b 32-b 33-b 34-b 35-b 36-c 37-c 38-c 39-b 40-b

Short Essay Questions

1. Explain the difference between bacteria and viruses.
 2. How does normal flora protect the human body?
 3. Explain how infection spreads in hospitals.
 4. Compare prokaryotic and eukaryotic cells.
 5. Compare bacteria, viruses, fungi, and parasites.
 6. Evaluate the importance of infection control in hospitals.
-

Bacterial Morphology

MCQ

1. A dormant bacterial structure shows no metabolic activity but can survive for years. This structure is:

- a. Capsule
- b. Endospore
- c. Plasmid
- d. Ribosome

2. A bacterium lacking capsule is more likely to:

- a. Increase virulence
- b. Resist phagocytosis
- c. Be more easily phagocytosed
- d. Form spores

3. The presence of teichoic acid in a bacterial cell wall suggests that the bacterium is:

- a. Gram-negative
- b. Gram-positive
- c. Virus
- d. Fungi

4. A bacterial cell shows an outer membrane and lipopolysaccharide (LPS). This indicates:

- a. Gram-positive bacteria
- b. Gram-negative bacteria
- c. Eukaryotic cell

d. Virus

5. Which of the following structures is essential for bacterial classification based on Gram stain?

- a. Capsule
- b. Cell wall
- c. Flagella
- d. Pili

6. Which structure is directly involved in determining the Gram reaction of bacteria?

- a. Ribosome
- b. Cytoplasm
- c. Peptidoglycan layer
- d. DNA

7. A bacterium with multiple flagella distributed over its surface is described as:

- a. Monotrichous
- b. Lophotrichous
- c. Peritrichous
- d. Amphitrichous

8. A bacterium lacking flagella is termed:

- a. Atrichous
- b. Monotrichous
- c. Peritrichous
- d. Amphitrichous

9. Which structure plays a major role in bacterial motility?

- a. Capsule
- b. Pili
- c. Flagella
- d. Cell wall

10. Pili differ from flagella in that pili are mainly involved in:

- a. Movement
- b. Energy production
- c. Attachment and conjugation
- d. Protection

11. Which structure is directly responsible for transfer of genetic material between bacteria?

- a. Capsule
- b. Flagella
- c. Sex pili
- d. Ribosome

12. A bacterial cell contains circular DNA not enclosed by a nuclear membrane. This DNA is located in:

- a. Nucleus
- b. Nucleoid
- c. Ribosome
- d. Capsule

13. Plasmids are important because they:

- a. Form cell wall
- b. Store proteins
- c. Replicate independently of chromosome
- d. Control shape

14. Transposons are best described as:

- a. Fixed DNA segments
- b. Mobile genetic elements
- c. Ribosomal units
- d. Structural proteins

15. Which structure is considered non-essential but contributes to virulence?

- a. Ribosome
- b. Cell membrane
- c. Capsule
- d. DNA

16. Which structure is always present in bacteria?

- a. Capsule
- b. Flagella
- c. Cell membrane
- d. Pili

17. The primary function of bacterial cell membrane is:

- a. Structural support
- b. Active transport and metabolism
- c. DNA storage
- d. Capsule formation

18. Oxidative phosphorylation in bacteria occurs in:

- a. Mitochondria
- b. Cytoplasm
- c. Cell membrane
- d. Nucleus

19. Mesosomes are associated with:

- a. Protein synthesis
- b. Cell division
- c. Capsule formation
- d. Movement

20. The cell wall is important for all EXCEPT:

- a. Maintaining shape
- b. Protection
- c. Protein synthesis
- d. Preventing osmotic lysis

21. Loss of cell wall may result in:

- a. Increased rigidity
- b. Cell lysis
- c. Increased DNA
- d. Increased movement

22. A bacterium exposed to unfavorable conditions forms spores. This process is called:

- a. Germination
- b. Sporulation
- c. Replication
- d. Conjugation

23. Germination of spores occurs when:

- a. Conditions are unfavorable
- b. Conditions become favorable
- c. Temperature decreases
- d. Nutrients decrease

24. Which of the following best explains why boiling does not kill spores?

- a. Spores lack DNA
- b. Spores are metabolically active
- c. Spores have thick protective layers
- d. Spores are small

25. The most reliable method to destroy bacterial spores is:

- a. Boiling
- b. Dry heat
- c. Autoclaving
- d. Filtration

26. Spores are medically important because they:

- a. Increase metabolism
- b. Are highly resistant
- c. Decrease infection
- d. Lack structure

27. Which bacterial component is primarily responsible for antigenicity?

- a. Cytoplasm
- b. Capsule
- c. Ribosome
- d. DNA

28. Lipopolysaccharide (LPS) is found in:

- a. Gram-positive bacteria
- b. Gram-negative bacteria
- c. Viruses
- d. Fungi

29. Which of the following contributes most to bacterial virulence?

- a. Ribosome
- b. Capsule
- c. DNA
- d. Cytoplasm

30. A bacterium with thick peptidoglycan and no outer membrane is classified as:

- a. Gram-negative
- b. Gram-positive
- c. Virus
- d. Protozoa

31. A bacterium with thin peptidoglycan and outer membrane is classified as:

- a. Gram-positive
- b. Gram-negative
- c. Fungi
- d. Parasite

32. Which structure is NOT involved in bacterial morphology classification (SSSMAC)?

- a. Size
- b. Shape
- c. Capsule
- d. Ribosome

33. Arrangement of bacteria refers to:

- a. Shape
- b. Grouping pattern
- c. Size
- d. Color

34. Which factor is most useful in rapid identification of bacteria in clinical labs?

- a. Size
- b. Gram stain
- c. DNA
- d. Movement

35. Which of the following best differentiates Gram-positive from Gram-negative bacteria clinically?

- a. Shape
- b. Size
- c. Cell wall composition
- d. Motility

36. Which bacterial structure is the main target for many antibiotics?

- a. Capsule
- b. Cell wall
- c. Ribosome
- d. Flagella

Key Answers

1-b 2-c 3-b 4-b 5-b 6-c 7-c 8-a 9-c 10-c
11-c 12-b 13-c 14-b 15-c 16-c 17-b 18-c 19-b 20-c
21-b 22-b 23-b 24-c 25-c 26-b 27-b 28-b 29-b 30-b
31-b 32-d 33-b 34-b 35-c 36-b

Scientific Terms Questions

1. A thick protective layer surrounding some bacteria that increases virulence.
2. A dormant, highly resistant structure formed under unfavorable conditions.
3. The region in bacterial cells where DNA is located without a nuclear membrane.
4. Extra-chromosomal circular DNA that can replicate independently.
5. Hair-like structures used for attachment and DNA transfer.
6. Long filamentous structures responsible for bacterial motility.
7. A staining technique used to differentiate bacteria into two major groups.
8. The main structural component of bacterial cell wall.
9. A component found in Gram-positive bacteria cell wall.
10. A component found in Gram-negative outer membrane responsible for toxicity.

Key Answers

1-Capsule 2-Endospore 3-Nucleoid 4-Plasmid 5-Pili
6-Flagella 7-Gram stain 8-Peptidoglycan 9-Teichoic acid
10-Lipopolysaccharide (LPS)

Bacterial Physiology

1. Bacterial growth is best defined as:

- a) Increase in cell size
- b) Increase in cell number
- c) Increase in metabolic activity only
- d) Increase in enzyme production

2. During bacterial growth, one bacterial cell forms a colony mainly due to:

- a) Cell enlargement
- b) Binary fission
- c) Budding
- d) Spore formation

3. The process by which bacteria reproduce asexually and produce two identical daughter cells is called:

- a) Mitosis
- b) Meiosis
- c) Binary fission
- d) Conjugation

4. In binary fission, the first step is:

- a) Cell separation
- b) DNA replication
- c) Septum formation
- d) Cytoplasm division

5. Generation time refers to:

- a) Time required for bacteria to die
- b) Time required for one cell to divide into two
- c) Duration of lag phase
- d) Time needed for spore formation

6. If the generation time of bacteria is short, the infection tends to be more severe because:

- a) Bacteria consume less nutrients
- b) Bacteria multiply rapidly
- c) Bacteria form spores quickly
- d) Bacteria become non-pathogenic

7. If a bacterium divides every 20 minutes, how many cells will be produced after 1 hour?

- a) 2
- b) 4
- c) 8
- d) 16

8. Which of the following is NOT a method used to detect bacterial growth in the body?

- a) Clinical signs
- b) Culture test
- c) Microscopic examination
- d) DNA replication

9. Which of the following is considered a clinical sign of bacterial infection?

- a) Increased oxygen level
- b) Fever and inflammation
- c) Decreased enzyme activity
- d) DNA mutation

10. Bacteria require all of the following for growth EXCEPT:

- a) Nutrients
- b) Proper environment
- c) Light
- d) Water

11. Which nutrient is mainly responsible for energy production in bacteria?

- a) Nitrogen
- b) Carbon
- c) Vitamins
- d) Minerals

12. Growth factors are:

- a) Inorganic salts
- b) Ready-made organic compounds
- c) Toxic substances
- d) Enzymes

13. Heterotrophic bacteria are characterized by:

- a) Ability to synthesize all nutrients
- b) Dependence on organic compounds
- c) No need for nutrients
- d) Ability to fix CO₂

14. Most pathogenic bacteria are:

- a) Autotrophs
- b) Heterotrophs
- c) Saprophytes
- d) Photosynthetic

15. Obligate aerobes require:

- a) Absence of oxygen
- b) High CO₂ only
- c) Presence of oxygen
- d) No gases

16. Obligate anaerobes grow best:

- a) In presence of oxygen
- b) In absence of oxygen
- c) In high CO₂ only
- d) In sunlight

17. Facultative anaerobes can grow:

- a) Only without oxygen
- b) Only with oxygen
- c) With or without oxygen
- d) Only in CO₂

18. Microaerophilic bacteria require:

- a) High oxygen levels
- b) No oxygen
- c) Low levels of oxygen
- d) Only CO₂

19. Capnophilic bacteria grow best in:

- a) High oxygen
- b) Low oxygen
- c) High CO₂
- d) No gases

20. Most pathogenic bacteria grow optimally at:

- a) 20°C
- b) 25°C
- c) 37°C
- d) 50°C

21. The optimal pH for most pathogenic bacteria is around:

- a) 2
- b) 5
- c) 7.2–7.6
- d) 10

22. Lack of moisture leads to:

- a) Increased bacterial growth
- b) Bacterial death or inhibition
- c) Faster metabolism
- d) Increased enzyme production

23. Bacterial enzymes are defined as:

- a) Structural proteins
- b) Biological catalysts
- c) Energy sources
- d) Genetic materials

24. The substance acted upon by an enzyme is called:

- a) Product
- b) Catalyst
- c) Substrate
- d) Inhibitor

25. Bacterial enzymes contribute to virulence by:

- a) Reducing metabolism
- b) Destroying host tissues
- c) Preventing growth
- d) Decreasing infection

26. Bacterial pigments are:

- a) Primary metabolites
- b) Structural components
- c) Secondary metabolites
- d) Enzymes

27. Exopigments are characterized by:

- a) Being inside the cell only
- b) Diffusing into the surrounding medium
- c) Being colorless
- d) Being toxic

28. Bacterial toxins are:

- a) Nutrients
- b) Harmless products
- c) Poisonous substances
- d) Structural proteins

29. Exotoxins are usually:

- a) Lipids
- b) Proteins
- c) Carbohydrates
- d) DNA

30. Endotoxins are mainly associated with:

- a) Gram-positive cell wall
- b) Gram-negative outer membrane
- c) Cytoplasm
- d) Ribosomes

Key Answers

1-b 2-b 3-c 4-b 5-b 6-b 7-c 8-d 9-b 10-c

11-b 12-b 13-b 14-b 15-c 16-b 17-c 18-c 19-c 20-c

21-c 22-b 23-b 24-c 25-b 26-c 27-b 28-c 29-b 30-b

Scientific terms

1. Bacteria that depend on organic compounds for energy and nutrition.
2. Bacteria that can synthesize their own food from carbon dioxide.
3. The process by which bacteria obtain energy in the presence of oxygen.
4. The process by which bacteria obtain energy in the absence of oxygen.
5. Bacteria that require oxygen for survival.
6. Bacteria that grow only in the absence of oxygen.
7. Bacteria that can grow with or without oxygen.
8. Bacteria that require low levels of oxygen for growth.
9. Bacteria that grow best in high carbon dioxide concentration.
10. Chemical substances required in small amounts for bacterial growth.

Key Answers

1-Heterotrophs 2-Autotrophs 3-Aerobic respiration 4-Anaerobic respiration 5-Obligate aerobes 6-Obligate anaerobes
7-Facultative anaerobes 8-Microaerophiles 9-Capnophiles

Normal Flora and Opportunistic infections

1. Which of the following microorganisms are considered normal flora in the human body?

- a) Only pathogenic bacteria
- b) Microorganisms regularly found on body surfaces
- c) Only viruses
- d) Only fungi

2. Which of the following is a characteristic of non-pathogenic microbes?

- a) Cause severe infections
- b) Always produce toxins
- c) Support vital body functions
- d) Require antibiotic treatment

3. Normal flora is mainly found in:

- a) Blood and CSF
- b) Internal sterile organs
- c) Skin and mucous membranes
- d) Brain tissue

4. Which of the following sites is normally sterile in healthy individuals?

- a) Skin
- b) Mouth
- c) Blood
- d) Intestine

5. Transient flora differs from resident flora in that it:

- a) Permanently colonizes the body
- b) Provides continuous protection
- c) Temporarily inhabits the body
- d) Is present from birth

6. Resident flora is characterized by:

- a) Temporary presence
- b) Instability
- c) Permanent colonization
- d) Harmful effects only

7. Colonization of normal flora begins mainly:

- a) Before birth in uterus
- b) During birth process
- c) At adolescence only
- d) After adulthood

8. The main function of normal flora in the skin is:

- a) Cause infection
- b) Produce toxins
- c) Form a protective barrier
- d) Destroy tissues

9. Which of the following is an example of opportunistic infection?

- a) Normal flora causing disease in healthy tissue
- b) Pathogens infecting sterile sites only
- c) Normal flora causing disease when host defenses are weakened
- d) Viruses infecting healthy individuals

10. Escherichia coli becomes pathogenic when it:

- a) Remains in intestine
- b) Produces vitamins
- c) Enters urinary tract
- d) Lives on skin

11. Which of the following best describes mutualism?

- a) One organism benefits, other harmed
- b) Both organisms benefit
- c) One benefits, other unaffected
- d) Both organisms harmed

12. Commensalism means:

- a) Both benefit
- b) One benefits, other unaffected
- c) Both harmed
- d) One harmed, other unaffected

13. Parasitism is defined as:

- a) Both organisms benefit
- b) One benefits, other harmed
- c) Both organisms unaffected
- d) Both organisms benefit equally

14. Which of the following is a function of normal flora?

- a) Cause tissue damage
- b) Prevent pathogen colonization
- c) Destroy immune system
- d) Reduce digestion

15. Competitive exclusion refers to:

- a) Killing host cells
- b) Occupying sites and preventing pathogen attachment
- c) Increasing pathogen growth
- d) Decreasing immunity

16. Which of the following is a cause of dysbiosis?

- a) Balanced diet
- b) Antibiotic therapy
- c) Normal immune function
- d) Proper hygiene

17. Dysbiosis is defined as:

- a) Normal microbial balance
- b) Absence of microbes
- c) Imbalance of normal flora
- d) Increase in beneficial bacteria only

18. Which nursing role helps maintain healthy normal flora?

- a) Excessive antibiotic use
- b) Poor hygiene
- c) Patient education and proper antibiotic use
- d) Ignoring infection control

Key answers

1-b 2-c 3-c 4-c 5-c 6-c 7-b 8-c 9-c 10-c
11-b 12-b 13-b 14-b 15-b 16-b 17-c 18-c

1. A patient received broad-spectrum antibiotics and developed diarrhea. The most likely cause is:

- a) Increased normal flora
- b) Dysbiosis
- c) Improved immunity
- d) Viral infection

2. A catheterized patient developed UTI caused by E. coli. This occurred because:

- a) E. coli is always pathogenic
- b) It is part of normal flora that became opportunistic
- c) It lives in blood normally
- d) It cannot survive outside intestine

3. A nurse did not perform hand hygiene between patients. This may lead to:

- a) Sterilization
- b) Cross infection
- c) Immunity
- d) Vaccination

4. A patient in ICU developed infection after 3 days of admission. This is:

- a) Community infection
- b) Nosocomial infection
- c) Viral infection
- d) Genetic disease

5. Which of the following situations increases the risk of opportunistic infection?

- a) Balanced flora
- b) Antibiotic overuse
- c) Proper hygiene

d) Strong immunity

6. The main role of nurses in preventing dysbiosis is:

- a) Overuse antibiotics
- b) Educate patients about drug use
- c) Ignore symptoms
- d) Reduce hygiene

key answers | 1-b | 2-b | 3-b | 4-b | 5-b | 6-b |

Short essay questions

1. Mention two functions of normal flora.
2. Define opportunistic infection.
3. Give two examples of sterile body sites.
4. What is dysbiosis?
5. Mention one role of nurses in preventing infection.

Sterilization and Disinfection

1. A nurse is preparing surgical instruments before an operation. What is the required process?

- a) Cleaning only
- b) Low-level disinfection
- c) High-level disinfection
- d) Sterilization

2. A patient has a wound that needs cleaning. Which agent should be used?

- a) Phenol
- b) Bleach
- c) Antiseptic
- d) Chlorine

3. A stethoscope used between patients should be processed by:

- a) Sterilization
- b) High-level disinfection
- c) Cleaning or low-level disinfection
- d) No cleaning needed

4. A catheter inserted into the bloodstream is classified as:

- a) Non-critical
- b) Semi-critical
- c) Critical
- d) Low-risk

5. Which of the following is an example of antiseptic?

- a) Chlorine
- b) Alcohol
- c) Phenol
- d) Formaldehyde

6. Which method is used for heat-sensitive liquids like vaccines?

- a) Autoclaving
- b) Dry heat
- c) Filtration
- d) Boiling

7. A nurse uses Betadine on a wound. This is an example of:

- a) Disinfection
- b) Sterilization
- c) Antisepsis
- d) Cleaning

8. Which item requires only low-level disinfection?

- a) Endoscope
- b) Surgical instruments
- c) Blood pressure cuff
- d) Catheter

9. Which sterilization method uses high-temperature steam under pressure?

- a) Dry heat oven
- b) Autoclave
- c) Filtration
- d) Radiation

10. Which of the following can survive disinfection and requires sterilization?

- a) Viruses
- b) Fungi
- c) Bacterial spores
- d) Protozoa

11. According to Spaulding classification, endoscopes are considered:

- a) Non-critical
- b) Semi-critical
- c) Critical
- d) Disposable

12. Which level of disinfection is required for semi-critical items?

- a) Low-level
- b) Intermediate-level
- c) High-level
- d) No disinfection

13. Which of the following is the first step before disinfection?

- a) Sterilization
- b) Cleaning
- c) Drying
- d) Packaging

14. A nurse is handling blood-contaminated material. Which is the best disinfectant?

- a) Alcohol
- b) Chlorine (bleach)
- c) Water
- d) Soap only

15. Which method is most effective in killing all microorganisms including spores?

- a) Boiling
- b) Disinfection
- c) Sterilization
- d) Cleaning

16. Which item requires sterilization?

- a) Bed sheets
- b) Surgical instruments
- c) Blood pressure cuff
- d) Thermometer

17. Which of the following is used as antiseptic on skin?

- a) Bleach
- b) Formaldehyde
- c) Alcohol
- d) Phenol

18. Which method is suitable for sterilizing surgical dressings?

- a) Filtration
- b) Autoclaving
- c) Radiation
- d) Boiling

19. Which of the following is considered non-critical item?

- a) Surgical blade
- b) Catheter
- c) Stethoscope
- d) Endoscope

20. Which of the following is the main goal of infection control in nursing?

- a) Increase infection
- b) Reduce patient care
- c) Prevent transmission of infection
- d) Avoid cleaning

Key Answers

1-d 2-c 3-c 4-c 5-b 6-c 7-c 8-c 9-b 10-c

11-b 12-c 13-b 14-b 15-c 16-b 17-c 18-b 19-c 20-c

Write the scientific term:

1. The complete destruction of all microorganisms including bacterial spores.
2. The elimination of most pathogenic microorganisms except bacterial spores.
3. The removal of organic matter and reduction of microorganisms from objects.
4. The use of chemical agents on living tissues to reduce microorganisms.
5. The process of making instruments safe to handle by removing pathogens.
6. Medical items that enter sterile tissues or vascular system.
7. Medical items that contact mucous membranes but do not penetrate sterile tissues.
8. Medical items that contact intact skin only.
9. The classification system used to determine level of disinfection or sterilization.
10. A method of sterilization using steam under pressure.
11. A sterilization method used for heat-sensitive liquids by removing microorganisms through a filter.
12. The process of using dry heat to destroy microorganisms.
13. Chemical substance used to destroy microorganisms on non-living objects.
14. Chemical substance safe for use on living tissues.
15. Highly resistant structures that require sterilization to be destroyed.

16. The process of destroying pathogens in blood spills using chlorine solution.

17. The method used to sterilize disposable medical equipment using radiation.

18. The step performed before disinfection to remove visible dirt.

19. The ability of microorganisms to resist disinfection methods.

20. The recommended method for sterilizing surgical instruments.

Key Answers

1-Sterilization 2-Disinfection 3-Cleaning 4-Antisepsis
5-Decontamination 6-Critical items 7-Semi-critical items
8-Non-critical items 9-Spaulling classification 10-Autoclaving
11-Filtration 12-Dry heat sterilization 13-Disinfectant
14-Antiseptic 15-Bacterial spores 16-Chlorine disinfection 17-
Radiation sterilization 18-Cleaning 19-Resistance 20-Autoclaving